CFD Analysis of Pulsatile Blood Flow through Bifurcating Artery and Investigating the Effect of Mechanical Properties

A project report is submitted to the Department of Mechanical Engineering, Khulna University of Engineering & Technology (KUET), and Bangladesh in partial fulfillment of the requirements for the Special Studies (ME 4000) course.

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DECLARATION

This is to certify that the thesis work entitled “CFD Analysis of Pulsatile Blood Flow through Bifurcating Artery and Investigating the Effect of Mechanical Properties” has been carried out by (Tonmoy Saha, Roll No. 1605056) in the Department of Mechanical Engineering, Khulna University of Engineering & Technology, Khulna, Bangladesh. The above work or any part of this work has not been submitted anywhere for the award of any degree or diploma.

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- Author

Abstract

Blood is a major component of the human circulatory system which is used as a working fluid in this present analysis of blood flow. For the diagnosis and treatment of cardiovascular and vascular diseases, understanding the dynamics of blood flow in arteries is very much important. The purposes of this present research is to develop a numerical simulation for clinical study to understand hemodynamics of blood and how it effects in bifurcating artery. This present research consists of three process which are pre-processing, numerical simulation and post-processing. In the pre-processing process a three-dimensional model was constructed by SOLIDWORKS. The finite element method (FEM) is used in order to simulate flow in a complicated geometry. The simulation is conducted to investigate hemodynamics of blood flow in left coronary artery of a human body. The aim of this study is to estimate the effect of blood flow through bifurcation artery. After simulation is carried wall shear stress, pressure and velocity is found. Maximum wall shear stress found for Newtonian and nonnewtonian blood model is 4.09 and 3.92 pascals. Maximum pressure found 42.5 and 35.9 Pa and finally maximum velocity found for both model is 20.7 and 21.9 m/s. Also, for model with plaque the result are found 51.94 and 49.14 pascals for wall shear stress, and the velocity found is similar for both models and finally maximum pressure for both models is 99.03 and 101.3 pascals. It can be said from this data that all models either Newtonian or non-Newtonian is almost similar and doesn't change much. But in vascular disease and atherosclerosis disease plaque plays important role. Because it is seen in change of models hemodynamic parameters.

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CHAPTER I

Introduction

1.1 Overview

Blood is a major component of the human circulatory system which is used as a working fluid in this present analysis of blood flow. Transportation of these substances in the circulatory system is being carried out using the plasma[1]. If a fluid obeys Newton's law of viscosity, it is said to be Newtonian. Plasma is a Newtonian fluid since it is mainly made up of water. The mechanical characteristics of blood are complicated. The rheological characteristics of blood are affected by interactions between blood cellular components. Blood can also be classified as non-Newtonian based on the size of blood vessels and blood flow behavior[2]. Blood has higher viscosity compared to plasma. Rise in hematocrit at very low shear rates leads to increase in viscosity of suspended particles and hence non-Newtonian behavior of blood is relevant[3]. A significant clinical location of atherosclerosis is the coronary artery system. The number of hemodynamic investigations in this system has been limited due to significant anatomic variation and a lack of thorough information on flow conditions in the coronary arteries[4]. The velocities of left coronary arteries have been measured by hot-film anemometry, pulsed Doppler ultrasonography [4-6] and laser Doppler anemometry [7]. Although the observations reveal significant flow properties, they are unable to provide comprehensive shear stress estimations. The continuous flow pattern has been qualitatively described using flow visualization in casts of coronary arteries [8, 9]. Laser Doppler anemometry (LDA) velocity measurements of simulated flow in casts of the human left anterior descending coronary arteries were reported by Mark et al. [10]. Plaque occurs mostly on the region of high curvature, complex geometry and bifurcation in arteries where shear stress of fluid and other fluid properties deviates from normal healthy artery[11]. Plaques were found to be closely related to WSS on the side branches. Atherosclerosis wreaks devastation on the curvatures and bifurcations of big and medium-sized arteries. There is a link between the genesis and advancement of cancer, according to several studies. which is in these zones, the illness is associated with a locally uneven flow field. Forces of fluid dynamics are intimately involved when it comes to Endothelial function. This is based on the Caro's idea of shear theory[12]

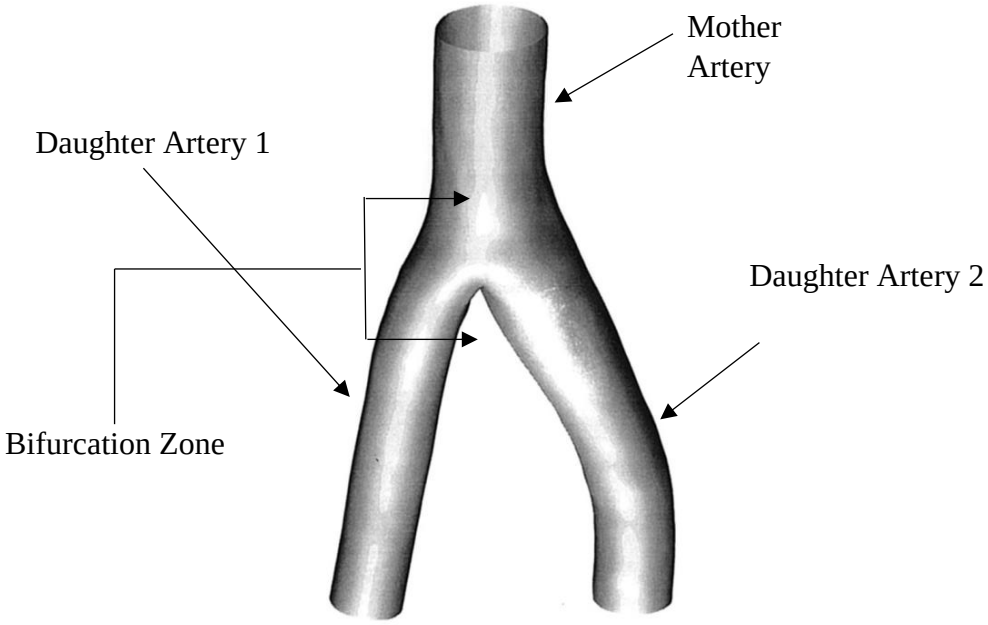


Fig. 1.1.1: A typical left coronary with bifurcation [13]

In figure 1.1.1 a typical left coronary artery is shown. Where left main stem is considered as mother tube and left circumflex and left anterior Descending is considered as daughter tube. A bifurcation which is can be defined as where left main stem get divided into two branches in this case LCx and LAD. Importance of determining the hemodynamics is that maximum wall shear stress occur at bifurcation point[14, 15]. Distribution of wall shear stress (WSS) in coronary arteries is regarded to be a crucial factor in the start of coronary heart disease, one of the leading causes of death. In Western society, there are a variety of reasons of death. Over the last few years, there has been debate on oscillating WSS also other hemodynamics properties has been linked to the emergence of atheroma Within the coronary arteries in particular whether this is the high, low, or medium reason[11, 16, 17]. There has been debate on whether high, low, or medium is the best option. oscillating WSS has been linked to the emergence of atheroma. In particular, within the coronary arteries atherosclerotic regions have been very much related with low and oscillating WSS hence hemodynamics [18-20]. Despite the clinical relevance of coronary artery atherosclerosis, research in this field has been limited due to the difficulty in collecting data on these arteries due to their small size and complicated geometries. The bifurcation of the left main coronary artery has been the main focus [21-24].

Despite the fact, the right coronary artery (RCA) has recently attracted some attention [18, 19, 25, 26].

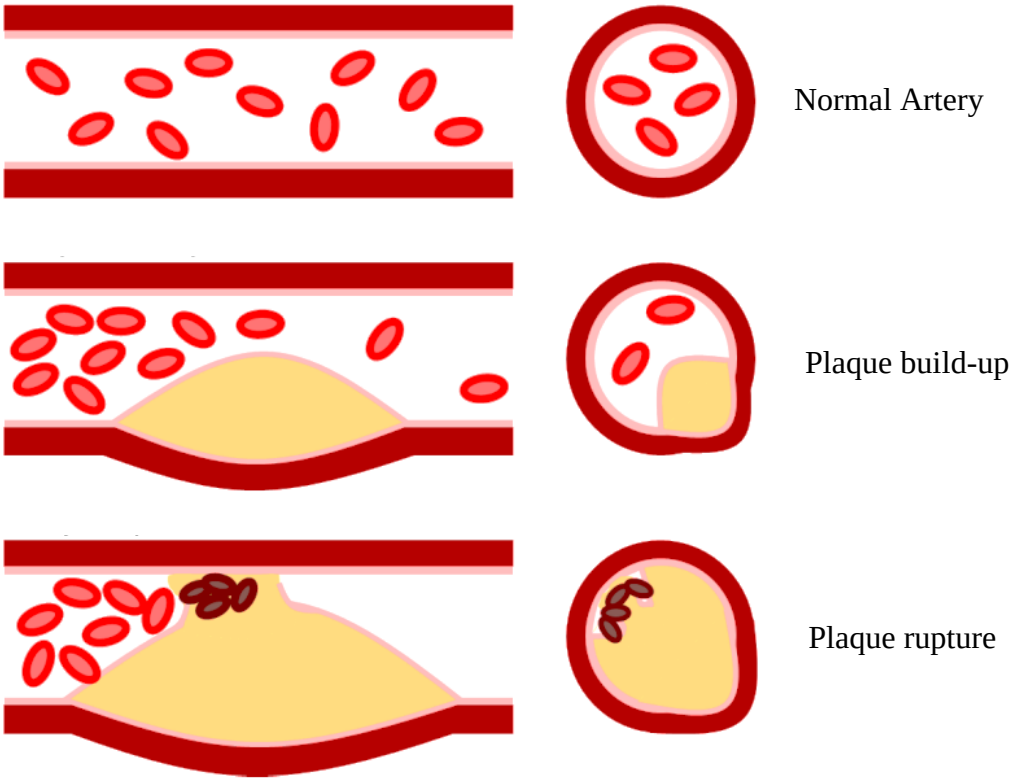


Fig. 1.1.2: Plaque development in the artery is progressing [27]

Atherosclerosis disease is related with plaque. Plaque mainly a blockage which disrupt the artery's blood flow. In figure 1.1.2 it is showed that a normal artery with normal blood flow (top), Atherosclerotic artery with plaque build-up and decreased blood flow (middle) atherosclerotic artery with a burst plaque (bottom) [27]. With these findings, understanding of the atherosclerosis development by exploring and quantifying the hemodynamic effect of coronary plaque sing CFD techniques is very much essential and it has been improved. The hemodynamic parameters of blood flow cannot be measured directly on blood vessels. Therefore, CFD techniques are employed to provide alternative ways for diagnosis of coronary artery disease[28]. To represent the viscous property of blood, there is no universally accepted model as the rheology of blood flow is complex. Some popular non-Newtonian models like Power-law, Casson, and Carneau models are used to simulate the blood behavior to different degrees of accuracy. In this study, a variety of non-Newtonian blood models are compared to

the standard Newtonian model in a realistic simulation of blood flow in the heart. These types of models are selected because, while there are several non-Newtonian models of blood rheology, none of them is universally accepted [29-31]. Also, wall shear stress, velocity profile and pressure are obtained through this study. This hemodynamic parameter will help to understand how plaque and complex geometry of artery causes in various vascular and atherosclerosis.

1.2 Objectives

The main objectives of this project are:

I. To investigate the effect of the artery wall on the local flow field using CFD analysis.

II. To investigate the wall shear stress, compare variety of non-Newtonian blood models with standard Newtonian model and flow disturbances in the formation and rupture of an atherosclerotic plaque using CFD analysis.

III. To identify the artery's geometric role in the appearance and progression of various vascular diseases and atherosclerosis disease and how mechanical properties affect this using CFD analysis.

IV. To analyze the hemodynamics of the bifurcation arteries of the human body using CFD analysis.

CHAPTER II

Literature Review

To measure coronary artery disease, Radhe, et al. investigated the influence of blood flow via the coronary artery in the human heart. Some of the major hemodynamic characteristics that may be measured non-invasively with computational fluid dynamics are velocity, wall shear stress (WSS), strain rate, and wall pressure distribution (CFD). These measurements are utilized to pinpoint the mechanical causes that cause plaque development and/or rupture in the left coronary arteries (LCA)[14]. Mahesh, et al. investigated blood flow in a bifurcating geometry having mother tube of diameter 5 mm and daughter tubes of 4 mm and 3 mm. At the end of the daughter tubes have been studied using CFD employing ANSYS Fluent 14.5. Simulations have been performed for non-Newtonian blood flow following the Carreau model with the pulsatile flow at the inlet to mimic blood flow conditions in the human body[15]. Hamidreza, et al. developed a workflow of a subject-specific CFD analysis using MRI to enhance estimating hemodynamics of the carotid artery. Time-of-flight MRI scans were used to construct three-dimensional computational models. Furthermore, different viscosity models of blood flow as a source of uncertainty were studied, using the axial velocity, wall shear stress, and oscillatory shear index, and a patient with carotid artery stenosis and its hemodynamic parameters were examined for developing geometry of artery[32]. Marie, et al. performed a numerical simulation system for the clinical study of a cerebral aneurysm which which consists of three processes: pre-processing, numerical simulation and post-processing. In the pre-processing process, the three-dimensional solid model was constructed from the computed tomography (CT) angiography raw data. The finite element method (FEM) is used in order to simulate flow in a complicated geometry. The simulation is conducted to investigate he-modynamics of blood flow in the carotid siphon under real flow conditions measured by Doppler ultrasound velocimetry technique. The results are visualized for a better understanding of the flow characteristics such as distributions of the flow pattern and the wall shear stress in the carotid siphon[33]. K. Perktold, et al. solved Pulsatile Non-Newtonian Blood Flow simulation through a Bifurcation with an Aneurysm numerically with the use of the finite element method. The results showed the disturbed blood flow in the bifurcation and the relatively low intra-aneurysmal flow circulation. In addition to the study of basic flow patterns in the segment, a comparison of non-Newtonian and Newtonian results was carried out. This comparison proved that for the considered large artery model under physiological flow conditions where the yield number is relatively low there is no essential difference in the results

[24]. Barbara M., et al. focused on blood flow through four different right coronary arteries. The wall shear stress in each of these arteries at a certain time in the cardiac cycle is studied using five non-Newtonian blood models as well as the standard Newtonian model of blood viscosity. The pattern of wall shear stress was found to be consistent across all models in the case of constant flow in a specific artery. The level of wall shear stress, on the other hand, is determined by the model employed[26]. A. J. Geers et, al. For two aneurysms, steady-state and transient models studied. The effects of a change in flow rate waveform were explored, and a virtual model was created. Treatment methods were used to compare post-treatment outcomes. Predictions of flow decrease**.** The impacts of a change in flow rate waveform were explored, and virtual Treatment methods were used to compare post-treatment outcomes to Predict flow reduction. It was discovered that the time-averaged wall shear stress on the aneurysm differed by less than 5%, and that the distribution of wall shear stress was qualitatively similar[34]. Wei et, al. studied to develop a model which is physiologically realistic in 3D CFD model. In this study, they investigate on left coronary artery under normal and disease conditions to estimate induced shear in blood flow stress, and then stimulate with the estimated shear stress. 3D geometry of left coronary artery was created in ProE and the simulation was done in Fluent and the stenosis condition was normal, 30%, 60% and 80%. Also, transient blood flow was used and turbulence model k-ω was used to create shear stress[35].

CHAPTER III

Methodology

3.1 Problem Statement

The problem under consideration is to investigate hemodynamics properties such as wall shear stress, velocity and pressure in a human artery and its effects on vascular diseases and atherosclerosis disease. In this study left coronary is chosen. This artery is chosen due to most vascular diseases occur on left coronary artery. A 3D model is designed in Solidworks 2020 and dimensions of the designed is taken from [14]. This is because artery dimensions are found only in CT and MRI. Due to limitations, Design was obtained from [14]. To numerically solve the problem Ansys Fluent 2020 was used. With the specific boundary condition this problem was solved.

3.2 Model

3.2.1 Bifurcation Artery model without plaque

Figure 3.2.1 shows a bifurcation artery which is divided into two branches when at section A,B,C and D has diameter of 6.6mm ,6.2mm ,5.3mm and 4.3mm [13].

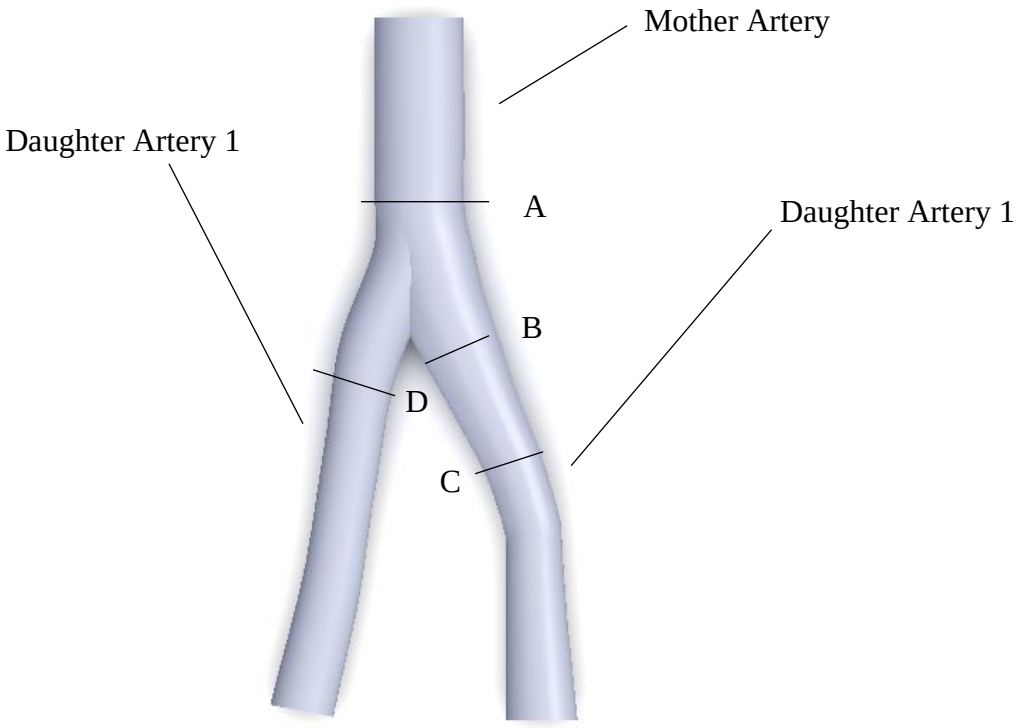


Fig. 3.2.1.1: 3D CT of a Bifurcation Artery

Botnar et al. Pulsatile flow velocity measurements were taken in a silicon rubber anatomically accurate copy of a human carotid artery bifurcation (true-to-scale elastic model). It's cast in a box that's with agarose gel with a concentration of 1.5 percent. This method of casting as a result, a model with almost solid walls emerges [13].

Table 1: Characteristic Length (mm) of geometric model [14]

Section	Diameter (mm)
A	6.6
B	6.2
C	5.3
D	4.3

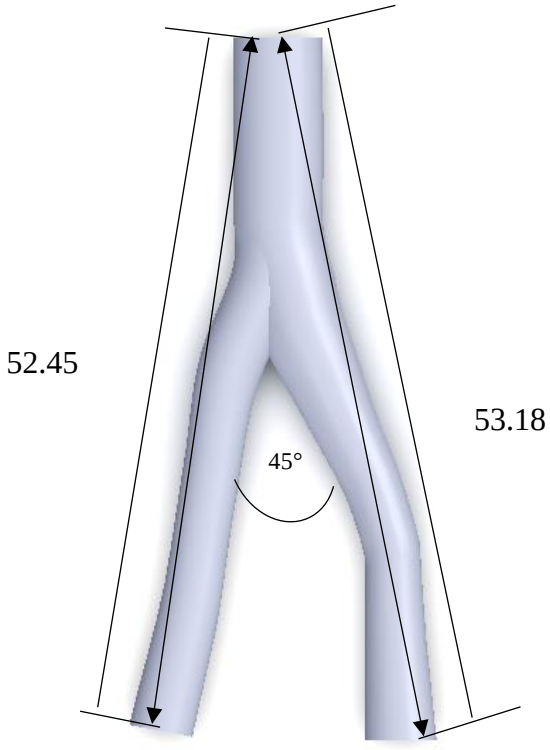


Fig. 3.2.1.2: Actual measurement of LCA

From figure 3.2.2 It is seen that bifurcation angle is 45 °. After gathering all data from [13] model of LCA is developed using software Solidworks 2020. Once the model is developed it is saved “IGS format” to use for further study.

3.2.2 Bifurcation Artery with Plaque

Figure 3.2.2.1 shows a bifurcation artery with plaque. This figure is similar with figure 3.2.1.1. It is modified version of figure 3.2.2.1. It can be seen that similarly there is a mother tube and two daughter tubes with same dimension as shown above. The main difference and main focus on this geometry is plaque. Usually, plaque formation does not look like this but mechanism of plaque is same which is to narrow the artery and disrupt the flow.

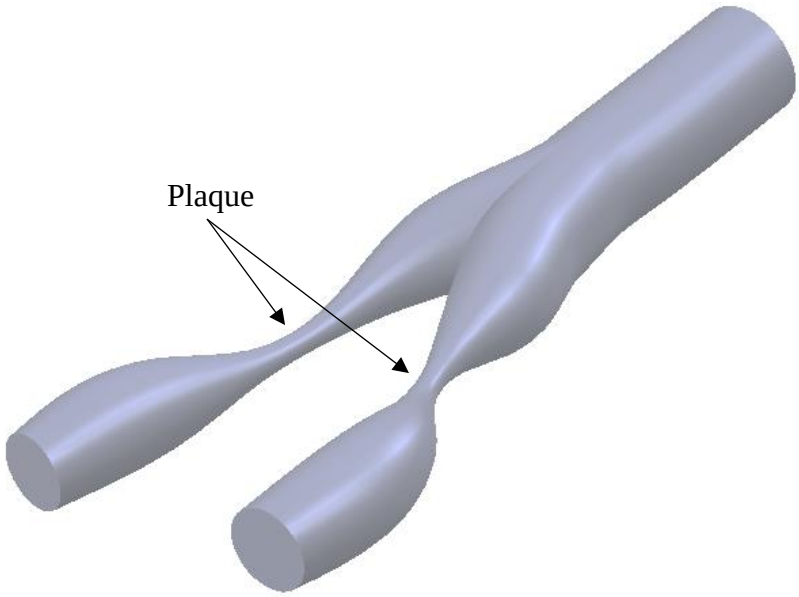
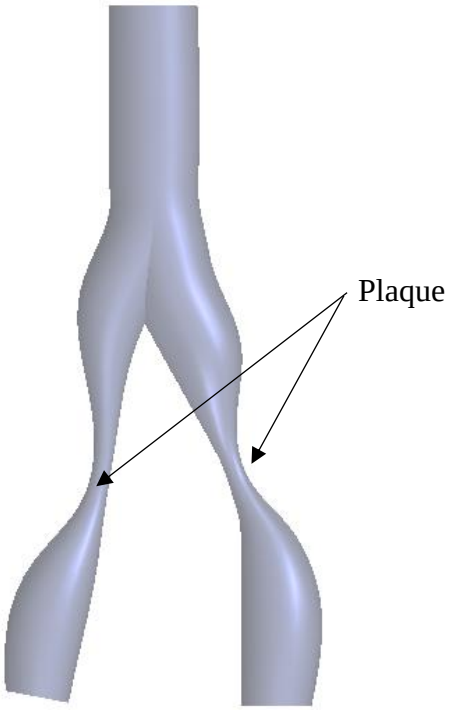
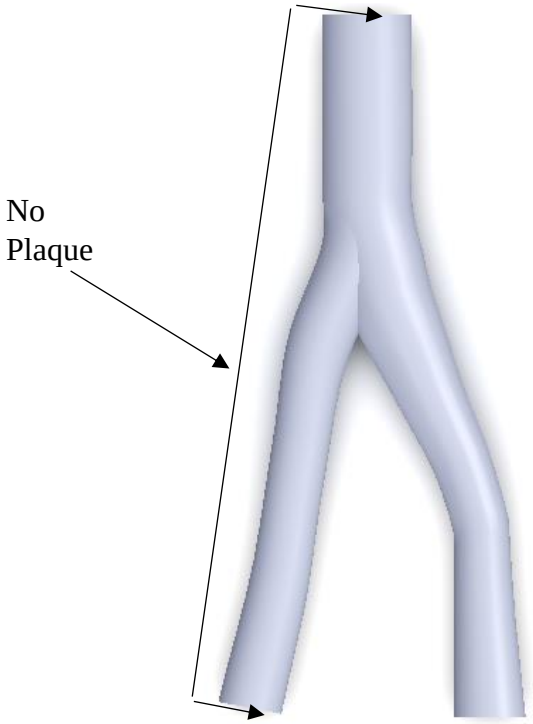


Fig. 3.2.2.1: Bifurcation artery with plaque (Isometric View)



(a)



(b)

Fig. 3.2.2.2: Comparison between artery (a) with plaque (b) without plaque

Figure 3.2.2.2 (a) shows a bifurcation artery with plaque. In the narrow zone of that artery is mainly shown as the plaque zone. Due to the plaque it is clearly seen that artery is deformed and in the narrow zone where plaque occurred flow will be disrupted and there will be more stress developed in the zone. In figure 3.2.2.2 (b) a normal artery is shown and there is no plaque zone. There is only bifurcation point in this model. So, there will normal flow through this artery. Comparison between this two model is discussed in details in result and discussion chapter.

3.3 Non-Newtonian blood models

There are two types of blood viscosity model Newtonian and Non-Newtonian. Also, There is various model for the non-Newtonian behavior of blood. Such models are Carreau Model, Walburb-Schneck model, Power Law model, Casson Model, Generalized Power law model. But, in this study, two blood viscosity models are considered which are Carreau model, Newtonian model. These models are used widely. They are all accepted by researchers because they are valid to simulate blood flow. These models parameter are discussed in Table 2. In Table 2 all parameters of blood models are contained also. These models are invented by various researchers [29, 36].

Table 2: Model in Blood Viscosity, μ, given in Poise (1 p = 0.1 Pa s) as a function of strain, $\dot{\gamma}$ given in s^{-1}

Blood Model	Effective Viscosity, μ
Newtonian Model	$\mu = 0.0345P$
Carreau model [26]	$\mu = \mu_\infty + (\mu_0 - \mu_\infty)[1+(\lambda\dot{\gamma})^2]^{(n-1)/2}$, $Where\ \dot{\gamma} = 3.313s,\ n = 0.3568, \mu_0 = 0.56P\ and\ \mu_\infty = 0.0345P$

3.4 Solution Methodology

It is considered that behavior of blood flow in LCA is governed by the Navier-Stokes equations.

$$\rho\left(\frac{\partial v}{\partial t}+V.\nabla V\right)=-\nabla\tau-\nabla P \qquad (1)$$

The continuity equation for an incompressible fluid:

$$\nabla.V=0 \qquad (2)$$

Where,

V is the 3D velocity vector

P pressure

ρ density

the shear stress term.

Writing the Navier stokes equations in this form allows the various non-Newtonian blood models, described above and given in Table 2, to be included. The governing equations are highly non-linear and must be solved using the techniques of computational fluid dynamics. To solve governing equations, a commercially available software package ANSYS fluent 2020 is used. Also, it is assumed that artery wall is rigid and the density if blood is $1060\ kgm^{-3}$.

3.5 Boundary Conditions

To solve the governing equations completely, a set of boundary conditions is required. Firstly, a no-slip condition is assumed along the walls of the artery. At the outlet, stress free conditions are applied and the gauge pressure is set to zero. At the inlet User-defined function (UDF) was used to implement the inlet, which simulates blood flow in the human body. Within each interval, the pulsatile profile is thought to comprise a mixture of two phases. The velocity at the intake fluctuates in a sinusoidal pattern during the systolic period. During the systolic phase, the sine wave has a maximum velocity of 0.5m/s and a minimum velocity of 0.1m/s. Each cycle lasts 0.5 seconds, assuming a heart rate of 120 beats per minute suggested this model for pulsatile blood flow[37]. Walls are considered as non-flexible and impermeable material also

no external forces is applied on it. Blood is assumed to be Newtonian and when non-Newtonian fluid, it is obeying the non-Newtonian fluid models discussed above.

3.6 Model Validation

To validate model of present study it was compared against the model of Nagargoje et al. [15]. A bifurcation artery with dimensions 0.005 m and 0.008m of mother tube and where daughter tubes having diameter of 0.004 m and 0.003 m at the outlet which linearly decreasing from bifurcation point. Also, the bifurcation angle was 45°. An exact model was created in SOLIDWORKS 2020 to validate the model of Nagargoje et al [15]. In the post processing a dimensionless number U/Umax vs R/Rmax is plotted to validate the model. Figure shows the comparison between two model Nagargoje et al. [15] and current model.

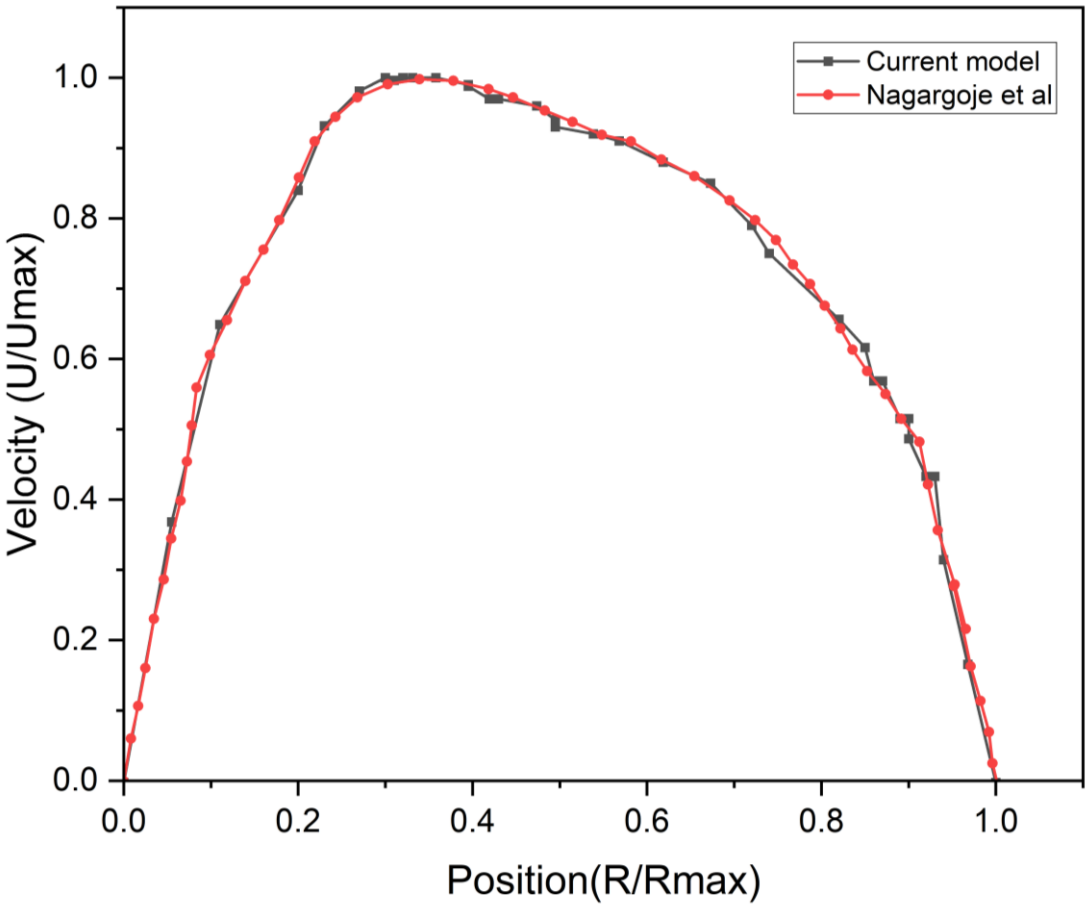


Fig. 3.6.1: Comparison between Nagargoje et al. [22] and Current model

In figure 3.6.1 red line denotes Nagargoje et al. [22] non dimensionless velocity vs position and black line denotes the current model non dimensionless velocity vs position where black line collides with red. But it is almost same as the red lines. Since, it is same for both model it can be said that model validation is done and the setup is selected for further study.

3.7 Mesh Independence Test

Mesh independence test was carried out using a series of simulation based on different number of elements for 2 models. By changing body size and inflation layer thickness it was achieved. This simulation was carried out for same setup that was used to do the validation. It can be seen from figure 3.7.1 wall shear stress varies with number of elements also it can be seen that meshes more than 2385633 gives accurate results. As a result, for further simulation the mesh with element number 2385633 is chosen for model without plaque.

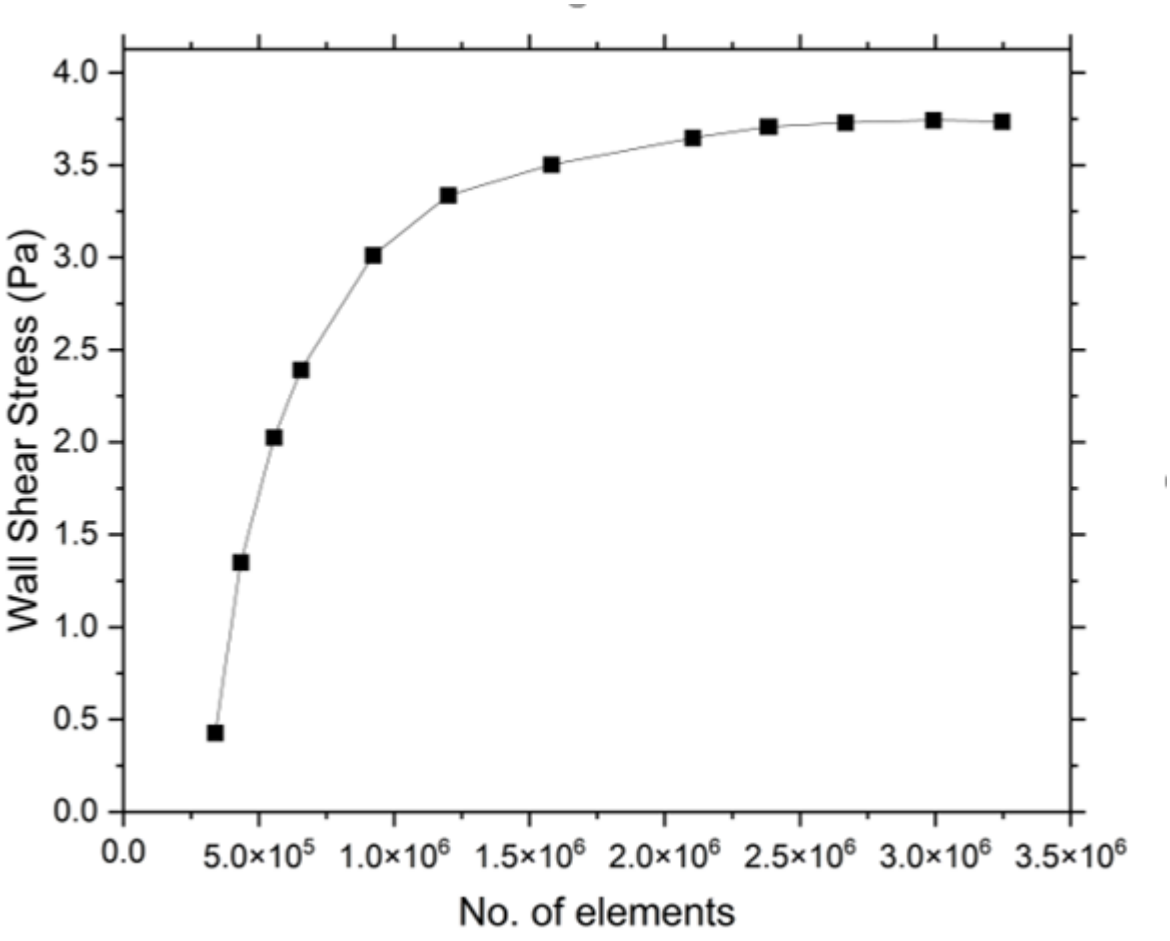


Fig. 3.7.1: Mesh independency test of model of artery without plaque

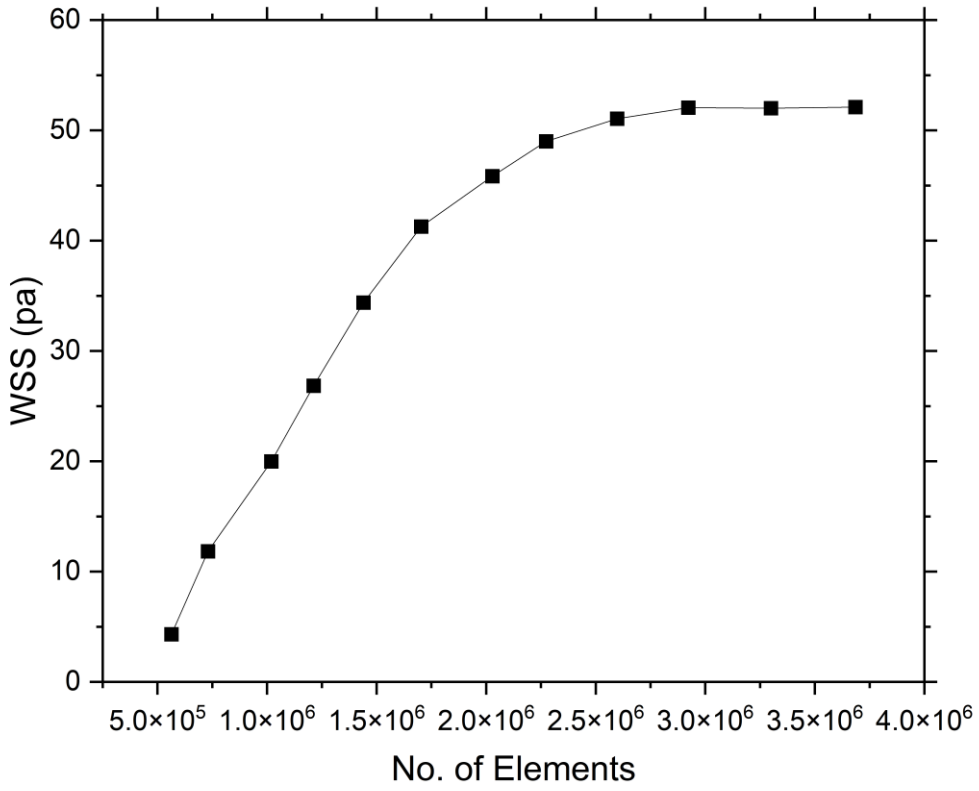
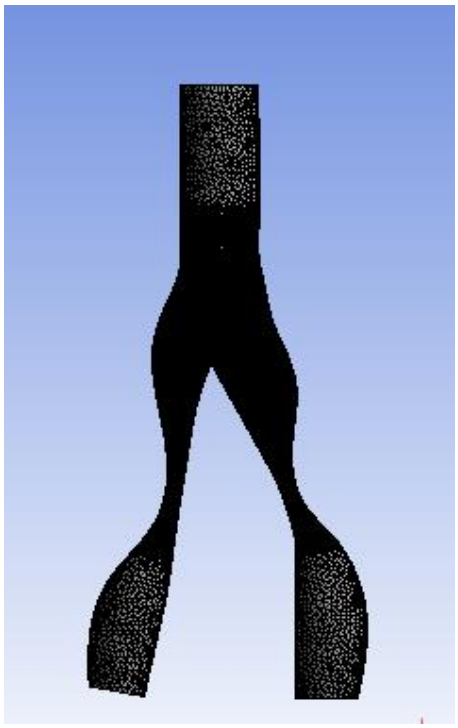


Fig. 3.7.2: Mesh independency test for model of artery with plaque

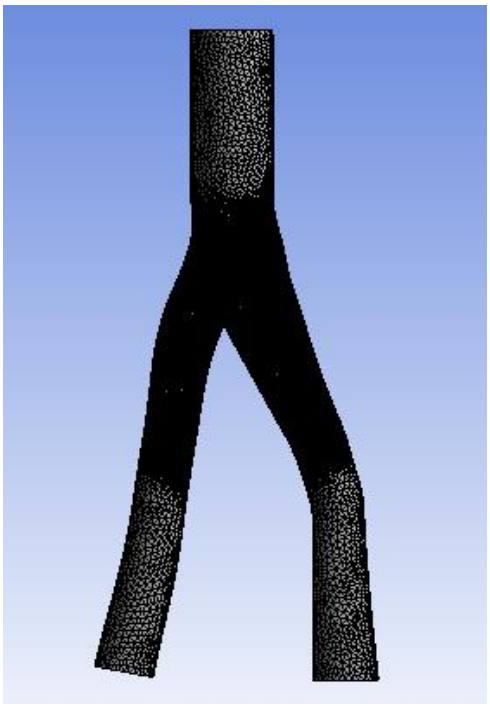
Mesh independency test for artery model with plaque has also gone through several simulation util a critical no. of elements is reached. It is seen that after 2922283 no. of elements there is no change in maximum wall shear stress. Further, if no. of elements is changed then there is no effect of maximum wall shear stress. So, as a no. of elements 2922283 is taken for simulation.

3.8 Mesh

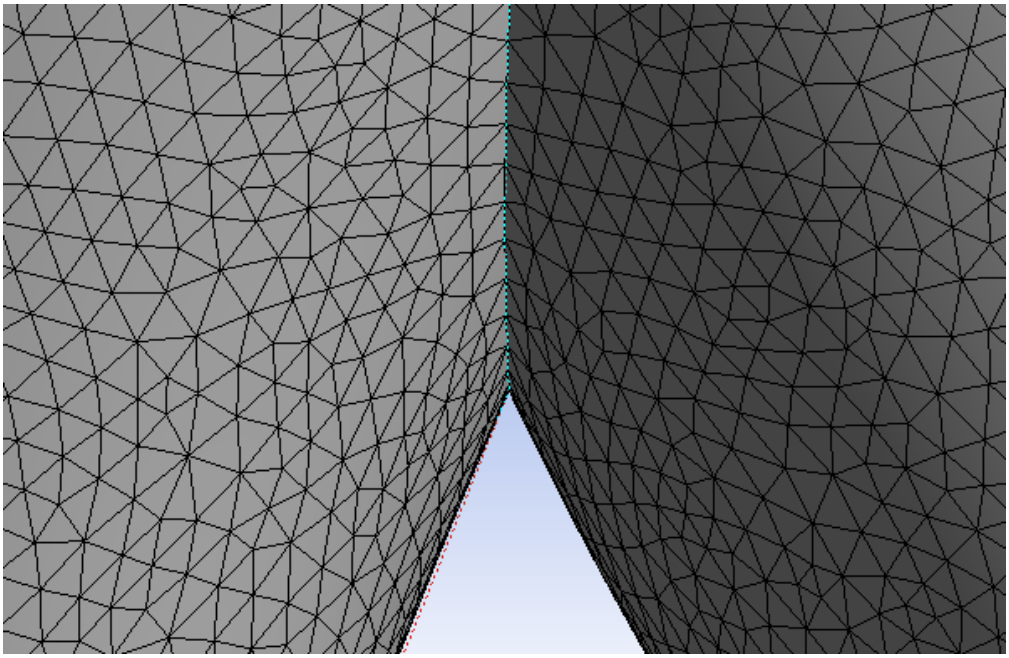
Since two types of models is introduced in this present study, each models have mesh quality and mesh structure. Since, models are all in three-dimensional structured mesh is difficult to achieve. So, non-structured mesh is generated for both models. Both models have body sizing of 0.5mm. In the bifurcation zone body sizing is decreased with sphere of influence method. This is because to obtain most accuracy in bifurcation zone. Also, an inflation layer with thickness of 1mm and 06mm for without plaque and with plaque is added. Both meshes figure is given below:



(a)



(b)



(c)

Fig. 3.8.1: Mesh of bifurcation artery

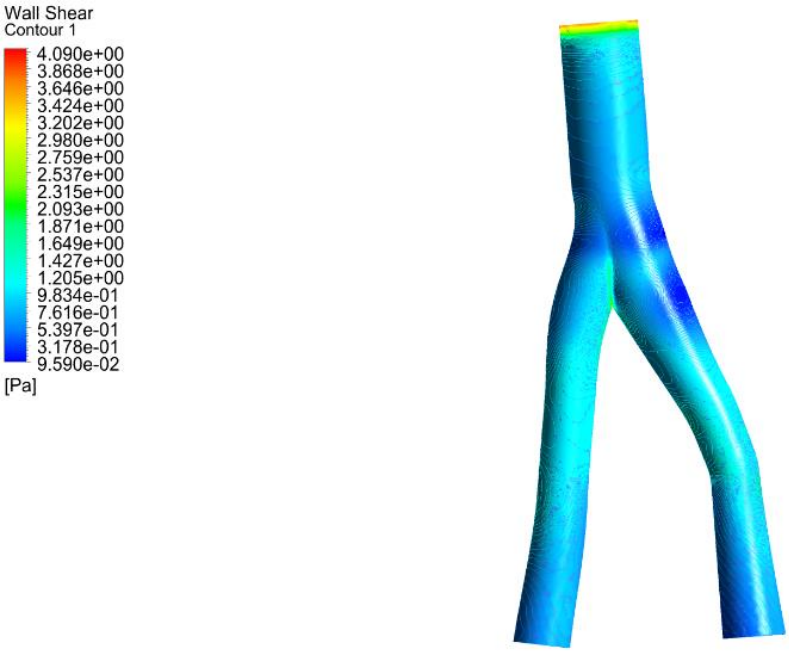
Figure 3.8.1 shows meshing of bifurcation of artery with plaque and without plaque. After meshing unstructured no. of elements found 2385633 and 2922283 for without plaque model correspondingly. This meshing is carried out for further simulation.

CHAPTER IV

Result and Discussion

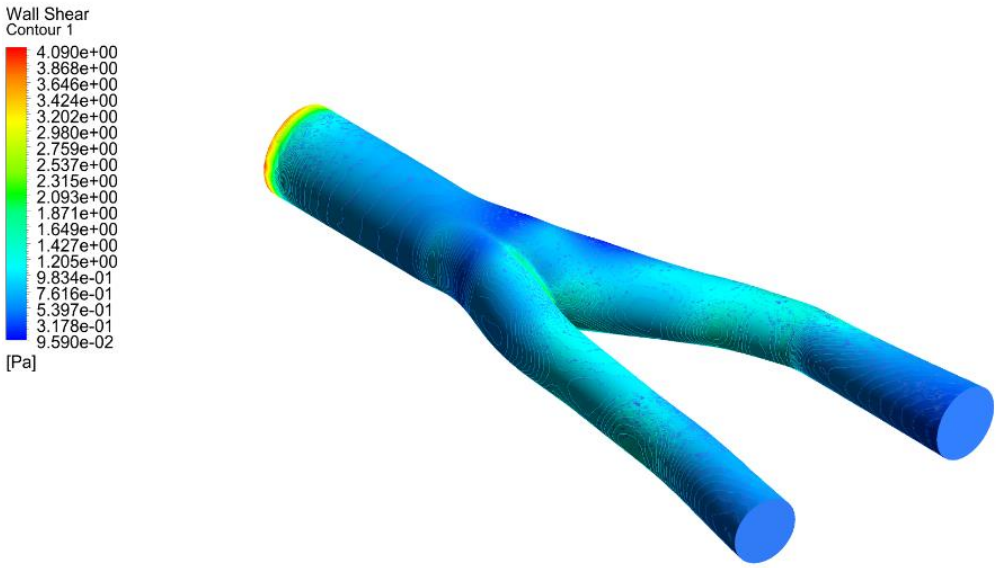
A series of simulation is done to obtained different parameters such as wall shear stress, pressure and velocity profile. These results are discussed below.

4.1 Changes of wall shear stress between non-Newtonian and Newtonian blood models without plaque



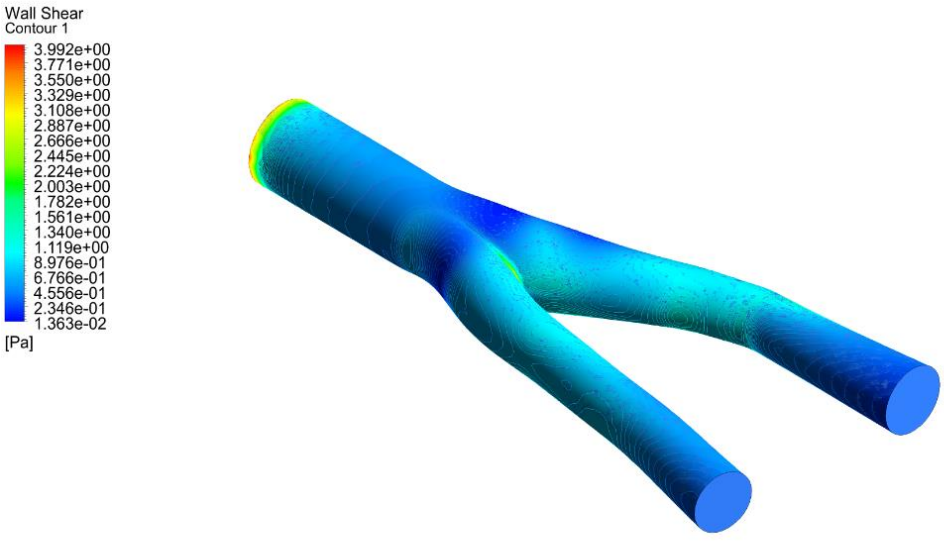
(a)

This contour of wall shear stress of non-newtonian blood models without plaque. Where maximu WSS developed is 4.09 pascals and minimum WSS developed is 0.0095 pascals. Also, from figure it is seen that maximum pressure occurs at bifurcation zone which is obvious and at wall there is shear stress is developed but compared to bifurcation zone it is not that significant. Importance of this contour is to know where maximum WSS develop and this zone is imporant for a artery.



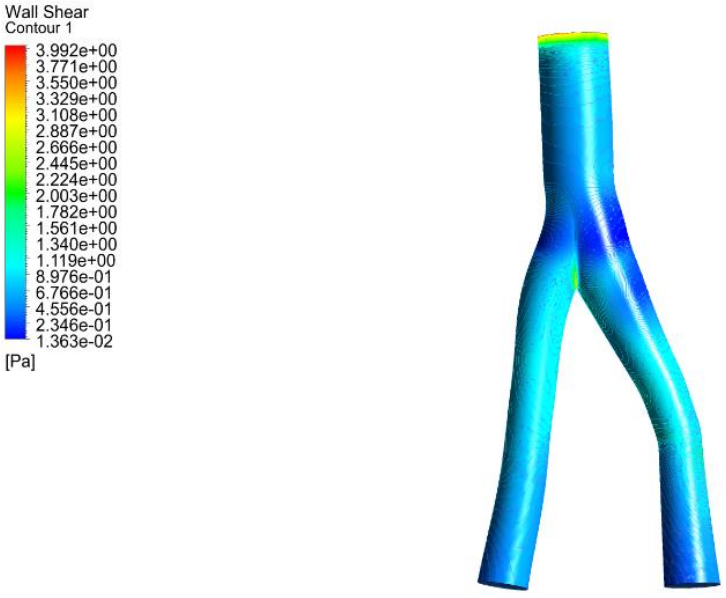
(b)

Fig. 4.1.1: Contour of wall shear stress (non-Newtonian) (a) Top View (b) Isometric View



(a)

This contour shows wall shear stress of Newtonian blood flow models without plaque and Maximum and minimum WSS found is 3.9 pascals and 0.013 pascals.



(b)

Fig. 4.1.2: Contour of wall shear stress (Newtonian) (a) Isometric View (b) Top View

In section 4.1 Contour of Wall shear stress between non-Newtonian and Newtonian blood models are compared and maximum pressure found is 4.09 and 3.92 pascals which doesn't differ much from each other. Also, it is seen that maximum shear stress occur at bifurcation zone. To know this closely wall shear stress vs position of wall is plotted which is shown in figure 4.1.3.

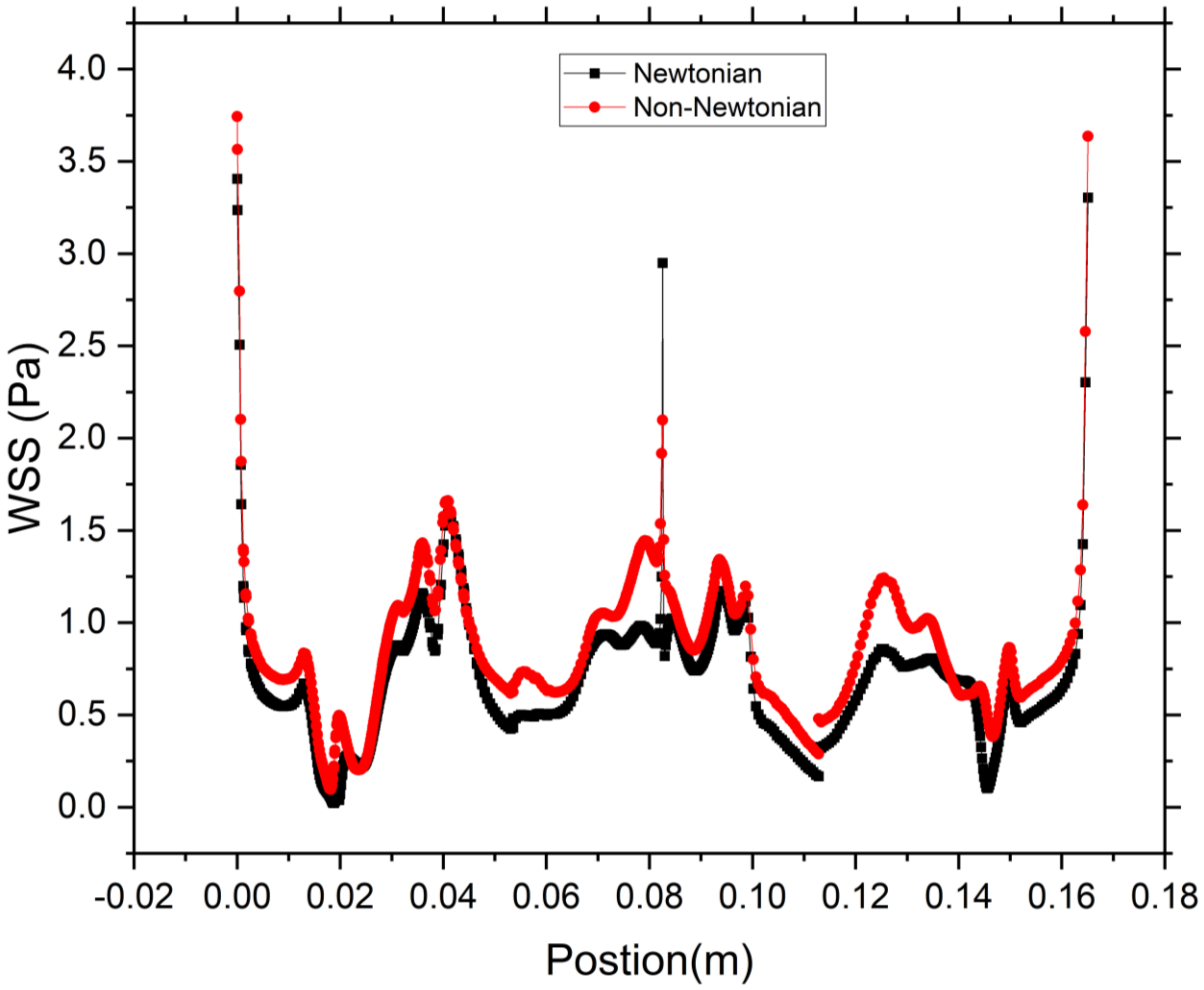
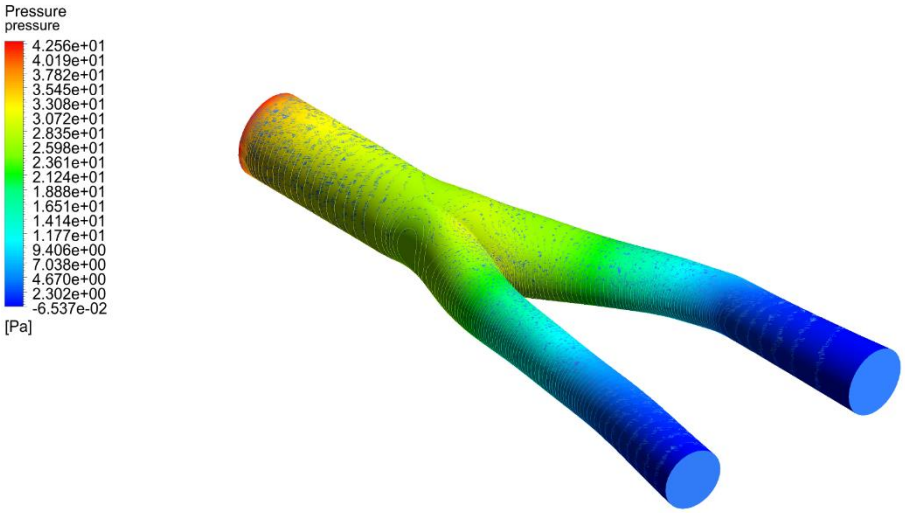


Fig. 4.1.3: WSS vs Position

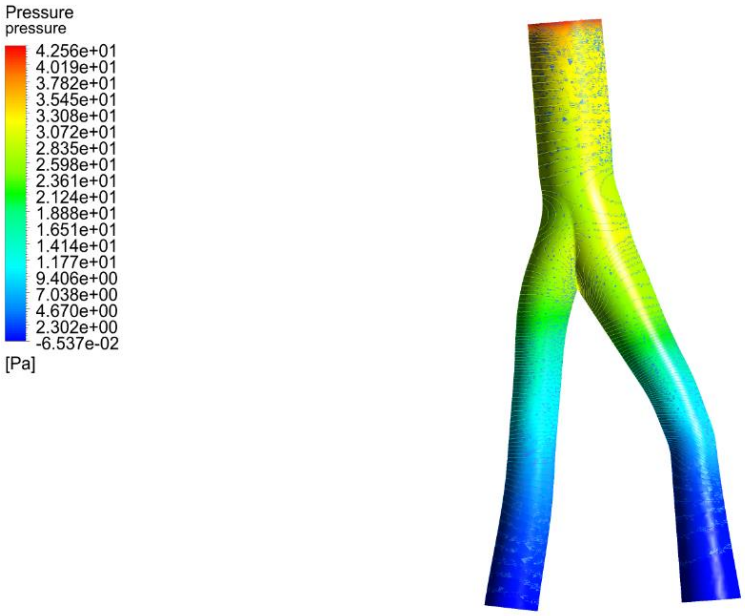
Figure 4.1.3 is mainly about comparing the differences between non-Newtonian and Newtonian blood models without plaque and finding that Newtonian model is valid or not. But it can be seen that these two models do not very much at each other. Also, other non-Newtonian models also do not very much with others. As it is seen that, their graph line is very similar. And their deviation is not that much. It is also seen from figure 4.1.3 that different blood models don't have so many differences at WSS. Their trend is almost same. Both curves differ by a small value.

4.2 Changes of Pressure between two blood models without plaque

From figure 4.2.1 and 4.2.2 pressure contour of non-Newtonians and Newtonian is given. And it is clearly seen that there is difference between maximum pressure between two models. Maximum pressure of non-Newtonian model is 42.5 Pa. On the other hand, maximum pressure of Newtonian model is 35.9 Pa. For better comparison a graph is given after the contour.



(a)

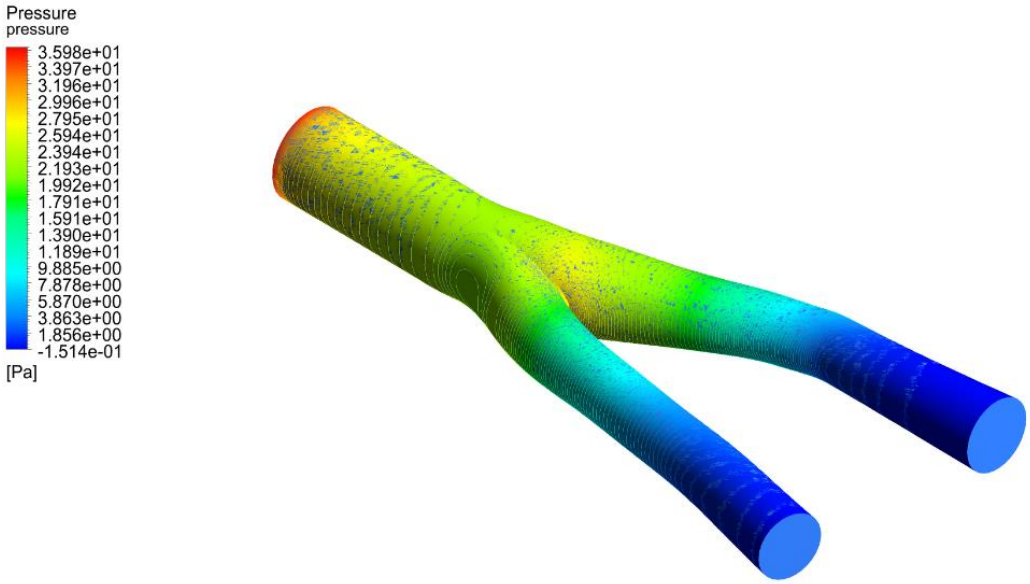


(b)

Fig. 4.2.1: Pressure Contour of Non-Newtonian Blood Models (a) Isometric View (b) Top view

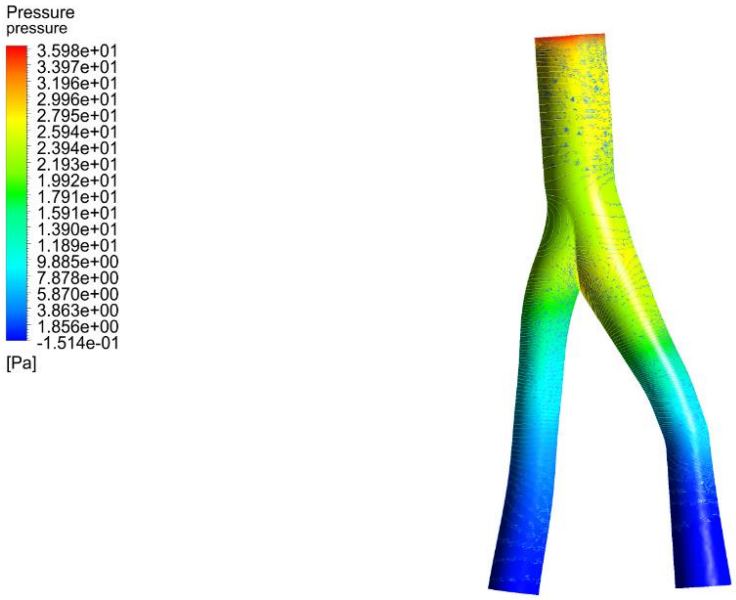
As WSS it is similar that in bifurcation zone maximum pressure occur for both model and this is the important. As maximum pressure is discussed minimum pressure should be discussed also. Minimum pressure found is that -0.6537 Pascal's

.



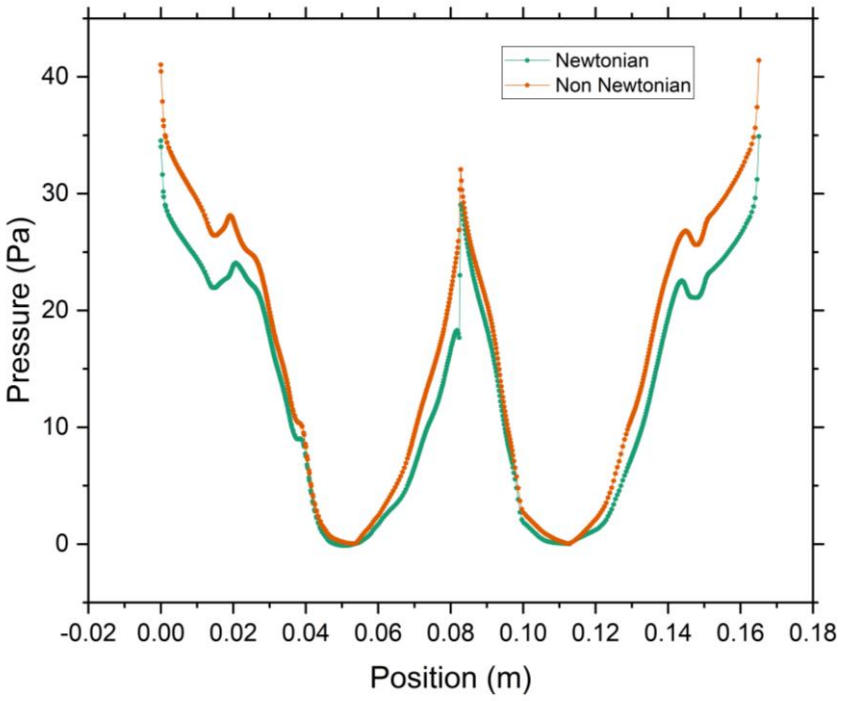
(a)

In this figure pressure for Newtonian model of blood is shown and the maximum pressure is found 35.98 Pascal's. As similar for non-Newtonian pressure is negligible. and it is mostly occurs at daughter tube 1 and 2. as maximum occurs at mother tube inlet but it is just for entry otherwise bifurcation zone is mostly important to consider.



(b)

Fig. 4.2.2: Pressure Contour of Newtonian Blood Models (a) Isometric View (b) Top view Fig.

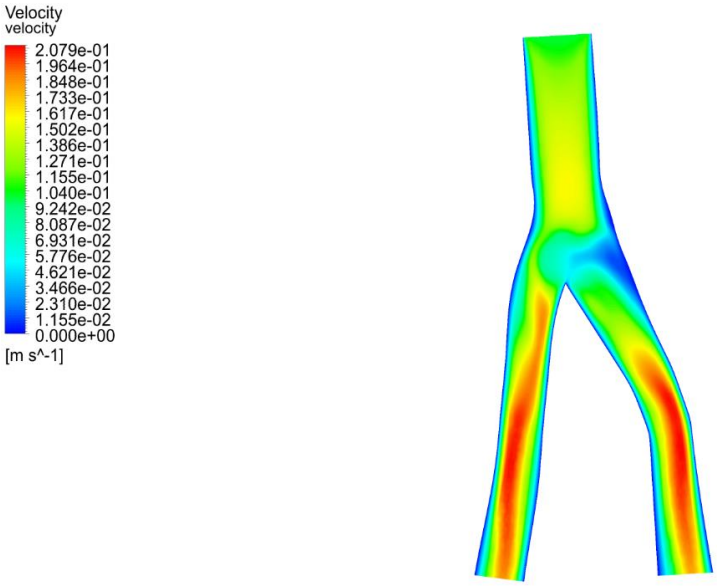


4.2.3: Pressure vs Position

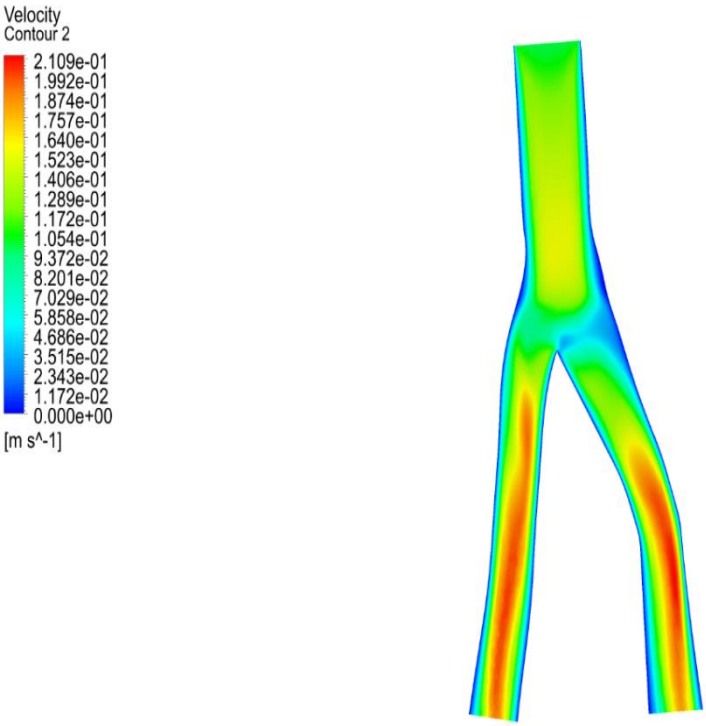
As said in 4.1.3 there was a similar trend, figure 4.2.3 has similar trend also. Also, it can be said there is not much difference in pressure.

4.3 Compression of velocity between two blood models without plaque

Velocity profile one of the vital hemodynamics parameters in blood flow. Figure 4.3.1 shows the contour of velocity profile and it is seen that despite of two different models the difference is almost negligible. Maximum velocity of carreau model is 20.7 m/s. On the other hand, Newtonian model's maximum velocity is 21.9 m/s.



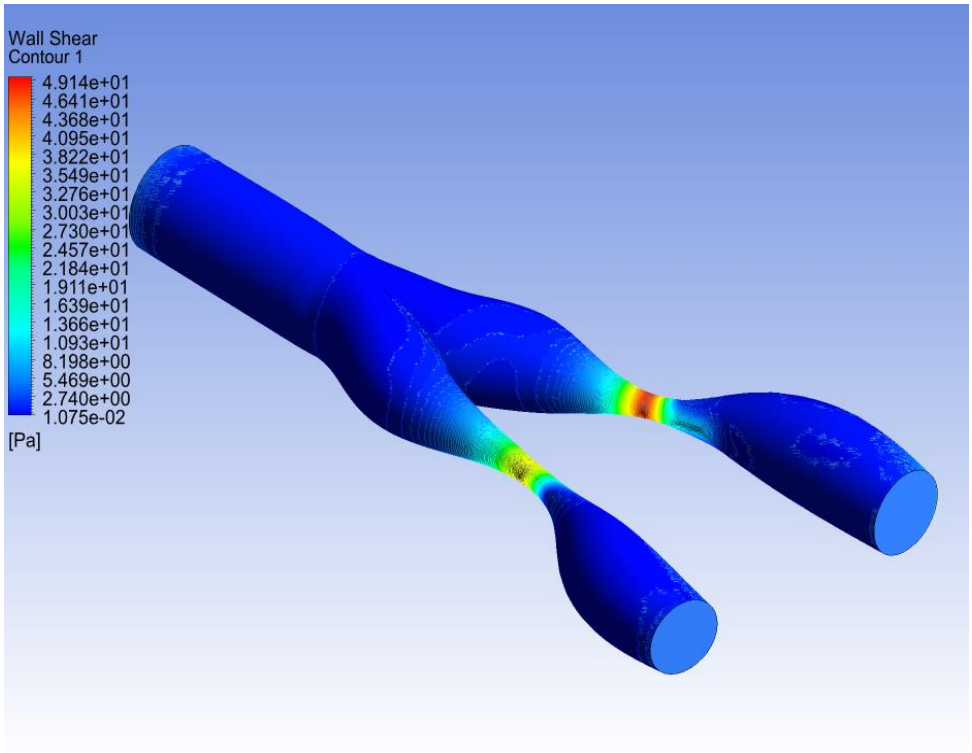
(a)



(b)

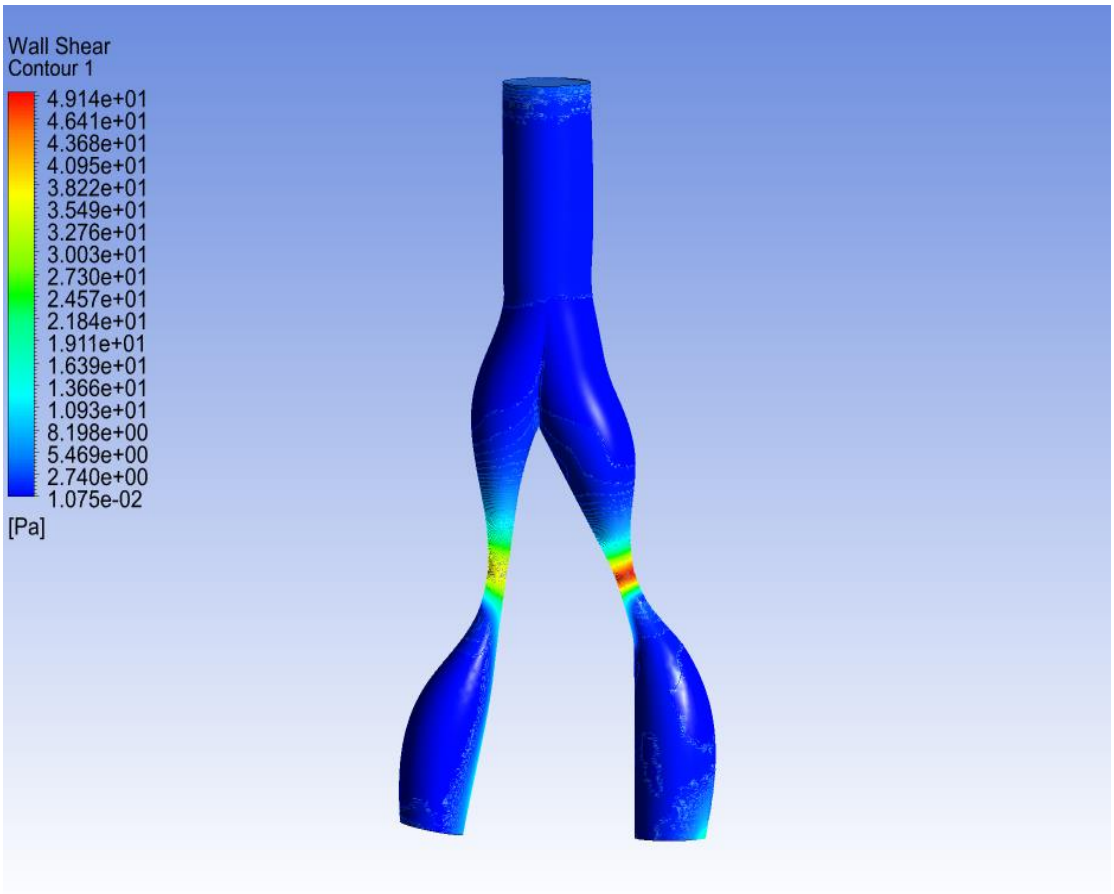
Fig. 4.3.1: Velocity Contour (a) non-Newtonian (b) Newtonian

4.4 Changes of wall shear stress between non-Newtonian and Newtonian blood models with plaque



(a)

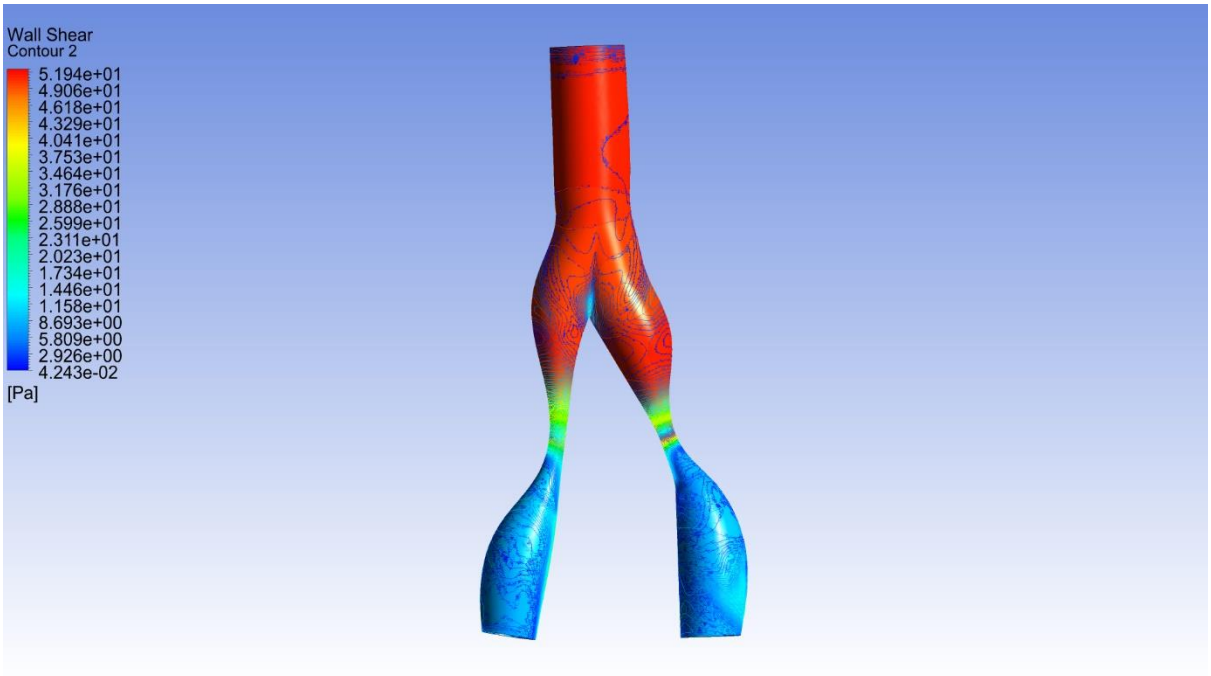
This figure shows the most important issues for vascular diseases. As it seen that most WSS developed in narrow zone which is due to plaque. There is no WSS in the wall compared to the bifurcation zone. This wall shear stress developed in this zone is most like to cause diseases and heart block and rupture of artery.



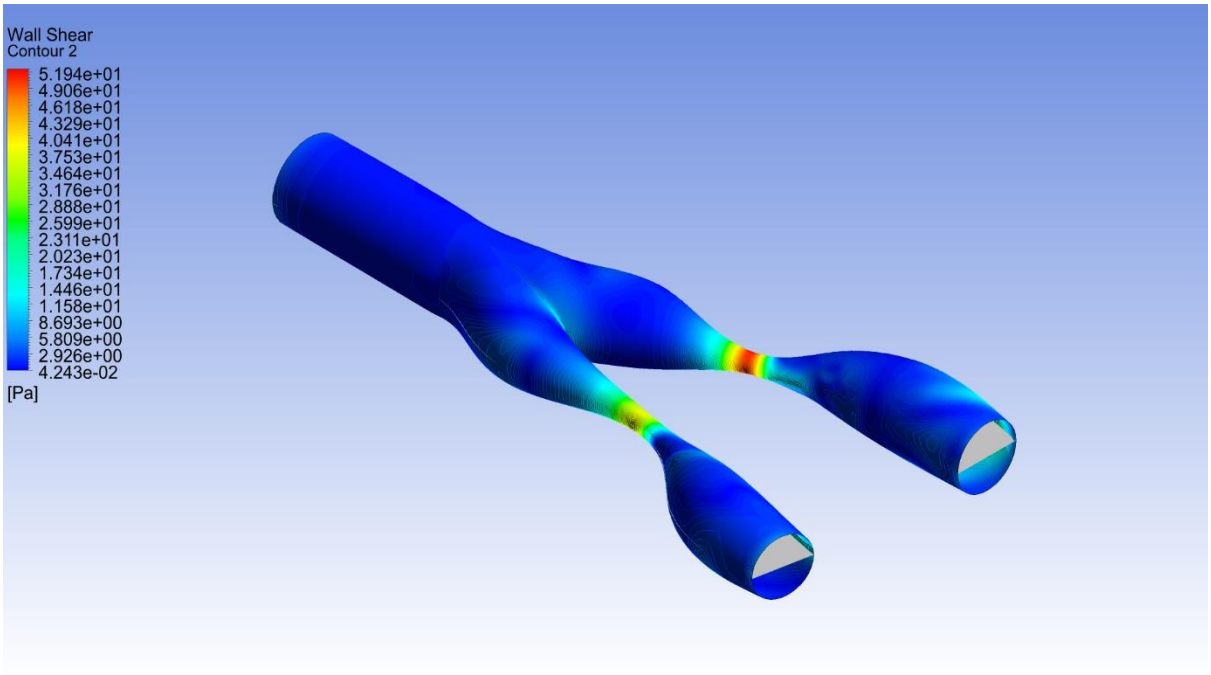
(b)

Fig. 4.4.1: Contour of wall shear tress(non-Newtonian) (a) Top View (b) Isometric View

Maximum wall stress for non-Newtonian blood model is found 49.14 Pascal's and minimum found 0.01. Also, between the both daughter tube one has more WSS and other has less. This because of the bifurcation. The flow of the blood is more in daughter tube and that has more WSS than others. This is happened because of the geometry.



(a)



(b)

Fig. 4.4.2: Contour of wall shear tress(non-Newtonian) (a) Top View (b) Isometric View

From figure 4.4.2 and 4.4.1 WSS can be compare. Maximum and Minimum WSS is found 49.14 and 51.94 pascals for model with plaque for non-Newtonian and Newtonian model. Also, a graph is given below for better understanding.

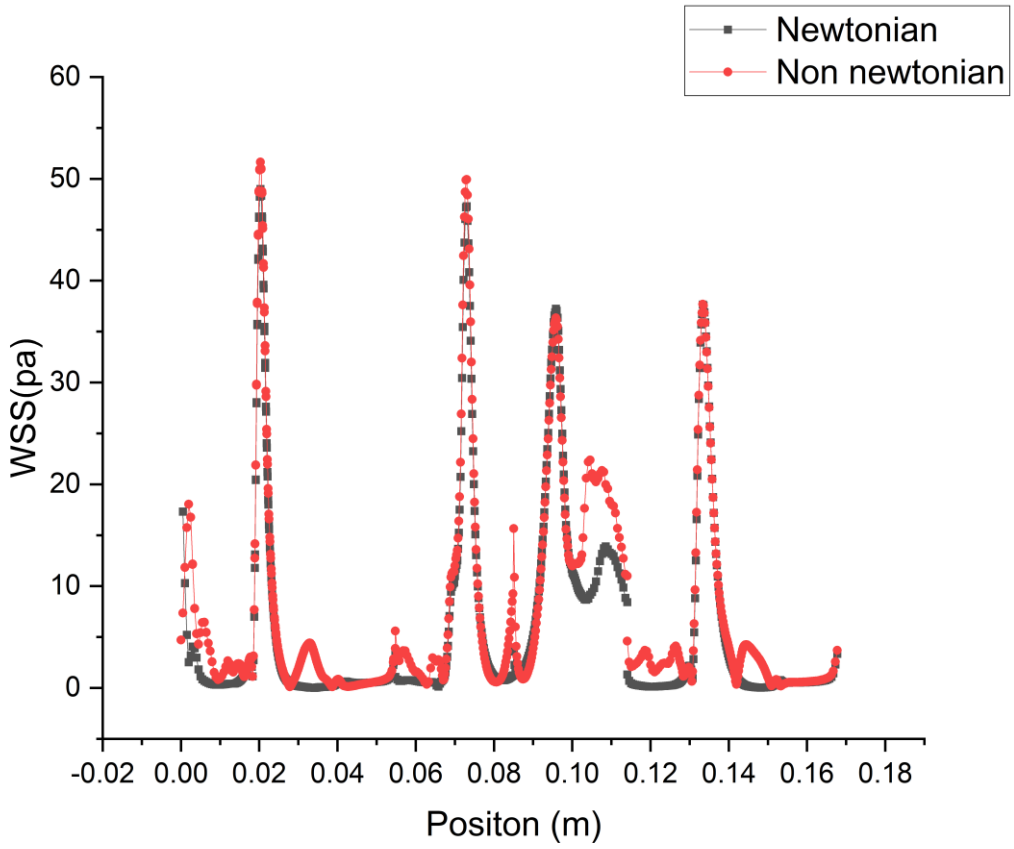
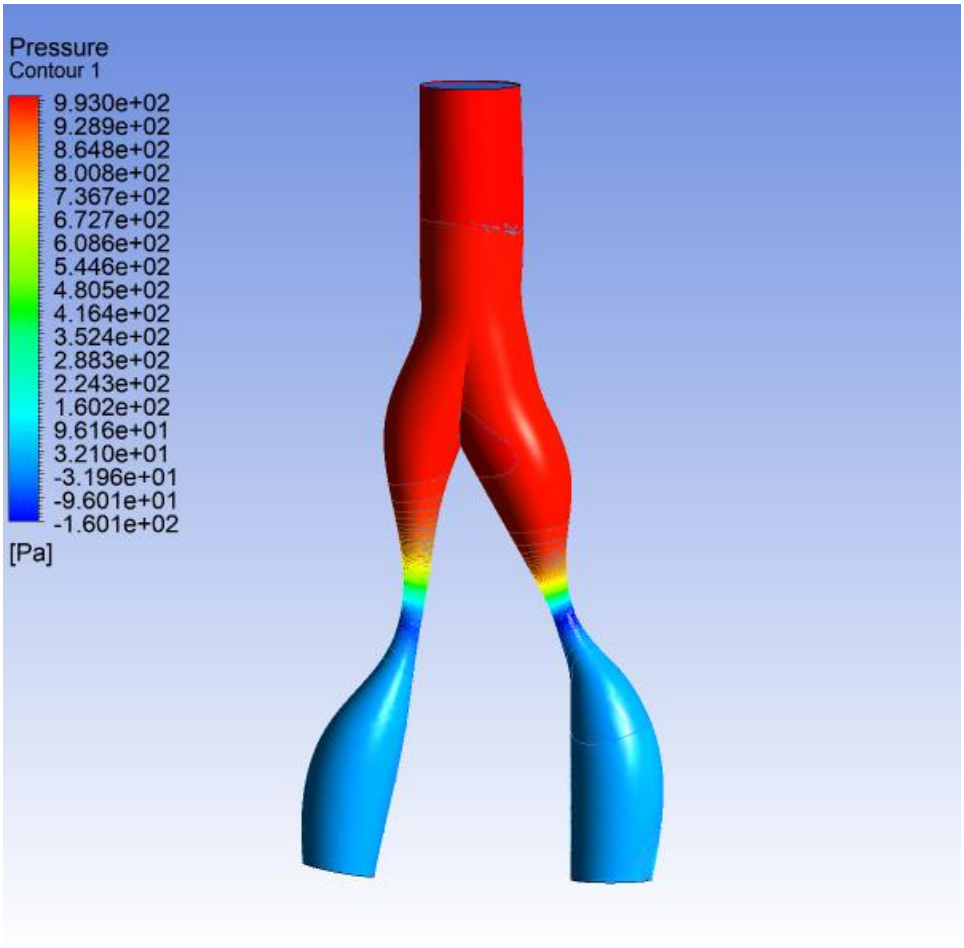


Fig. 4.4.3: WSS vs Position

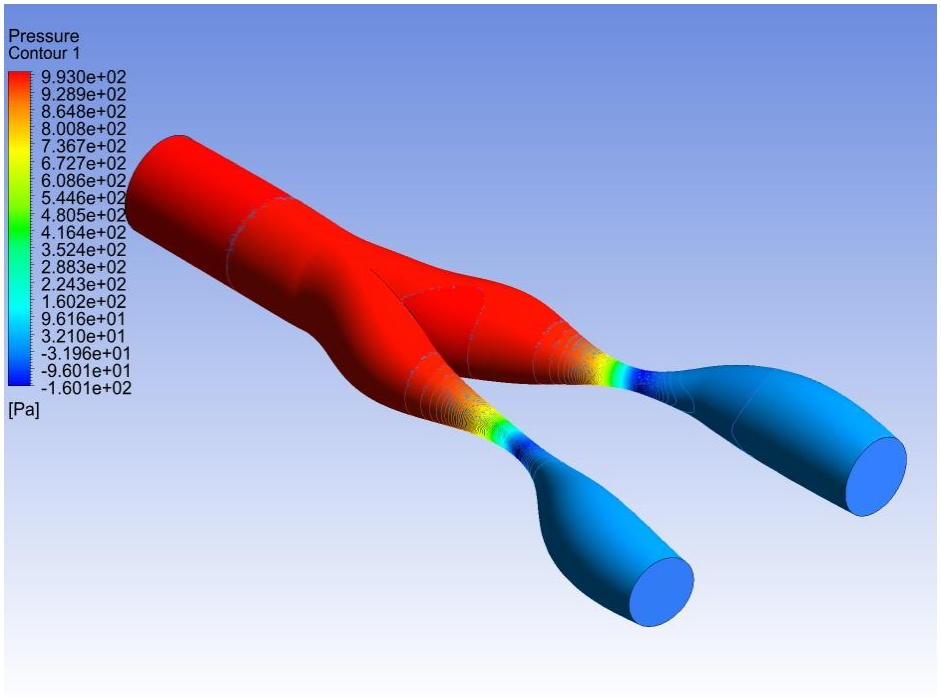
Figure 4.4.3 shows wall shear stress vs position superposition graph and it is seen that both graph has similar trends and also collide with one and other. So, it can be said that blood models could be a lot. But their differences are not that much.

4.5 Changes of Pressure between two blood models with plaque



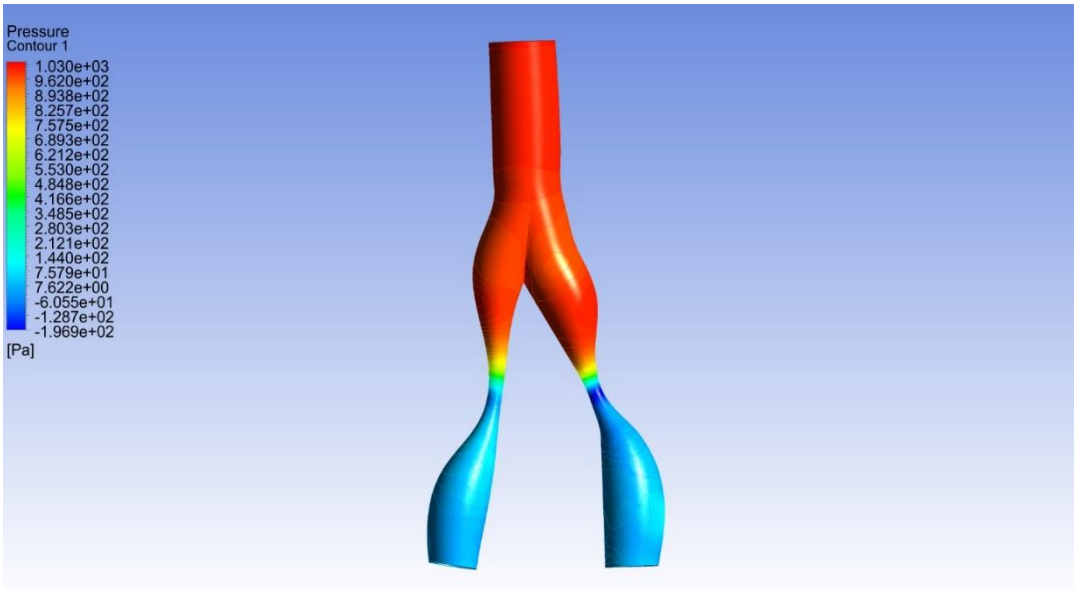
(a)

It is a top view of bifurcation artery with plaque and maximum pressure found is 99.3 pa. It can also be seen that all pressure is developed before bifurcation and no pressure is found after that. It is clearly understandable this is because of the plaque and narrow artery there.

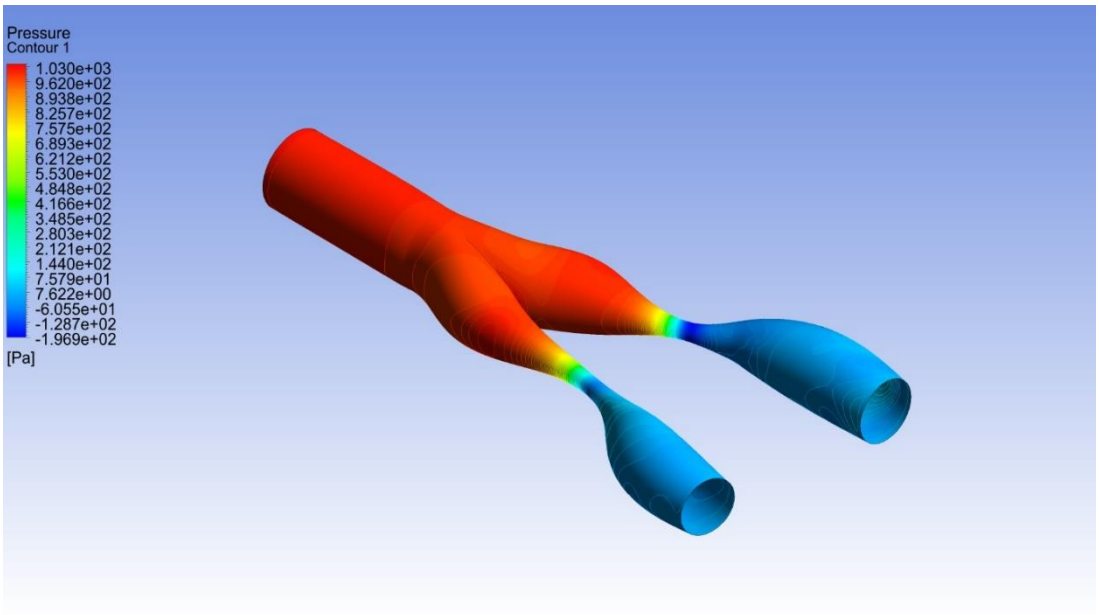


(b)

Fig. 4.5.1: Pressure Contour of Non-Newtonian Blood Models (a) Isometric View (b) Top view



(a)



(b)

Fig. 4.5.2: Pressure Contour of Newtonian Blood Models (b) Isometric View (a) Top view

This is a Newtonian blood models with plaque. and similar like non Newtonian blood models pressure is before the plaque zone. After that pressure is negligible compared to before plaque. It proves that plaque could be dangerous for heart blockage and create extra pressure than normal.

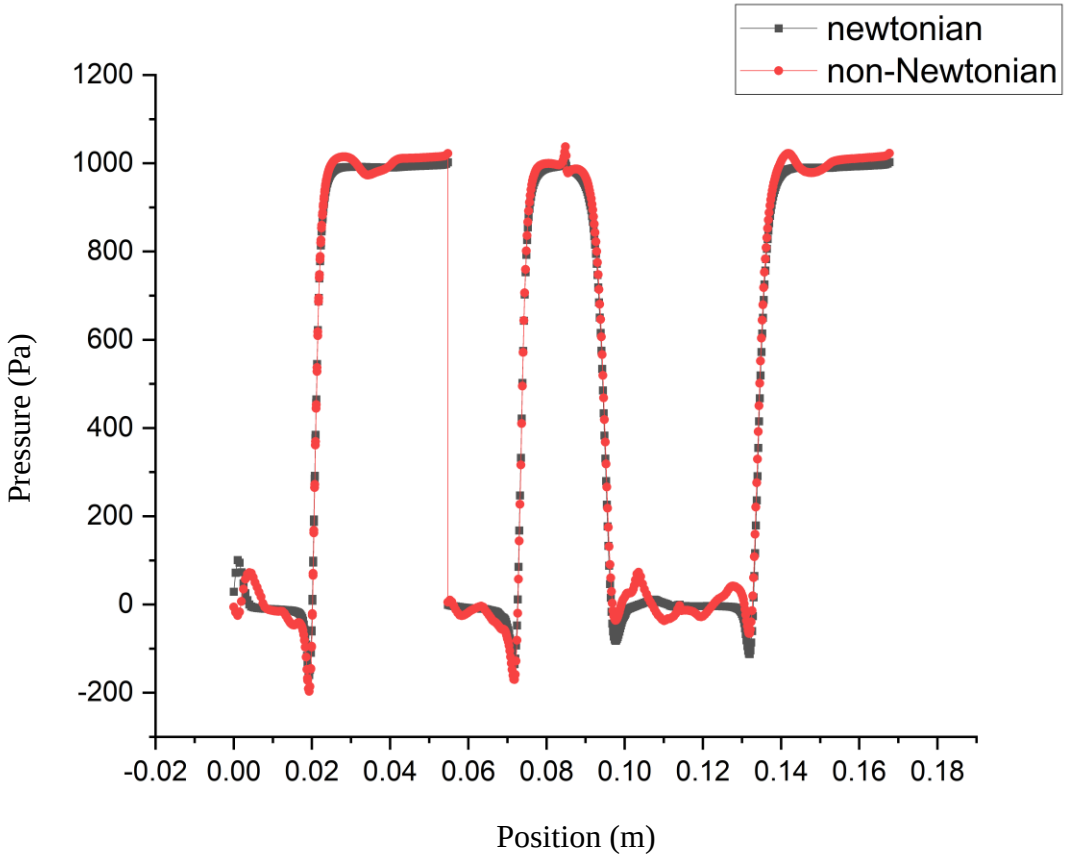
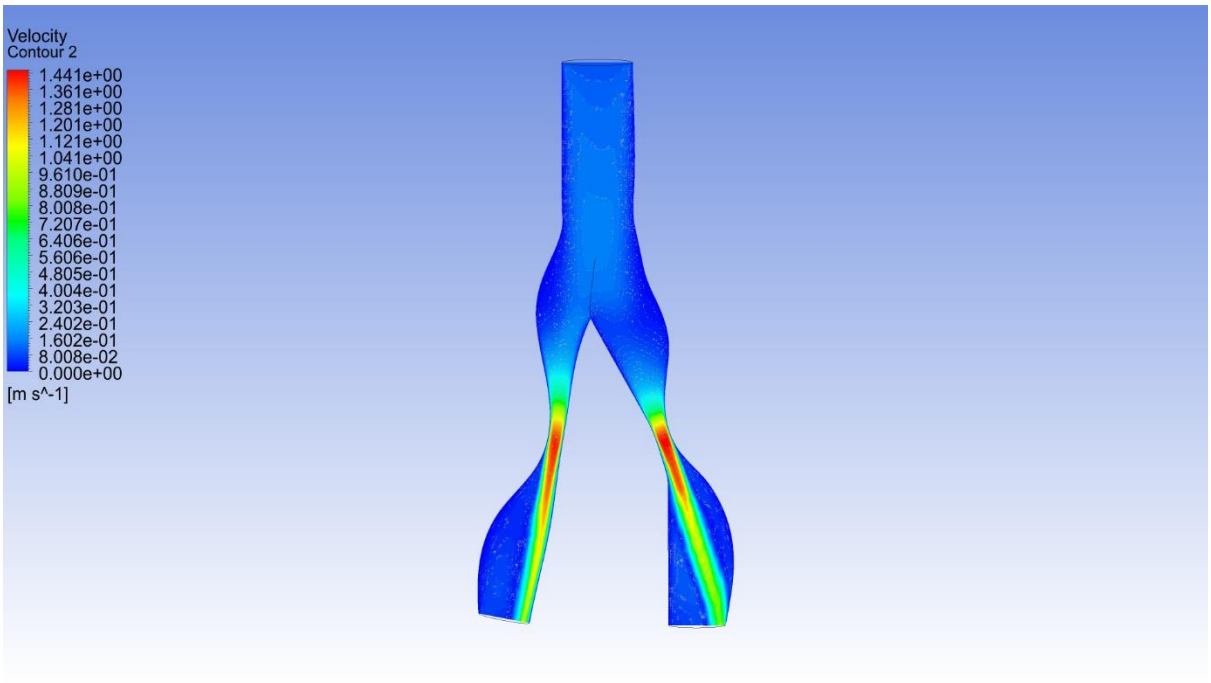


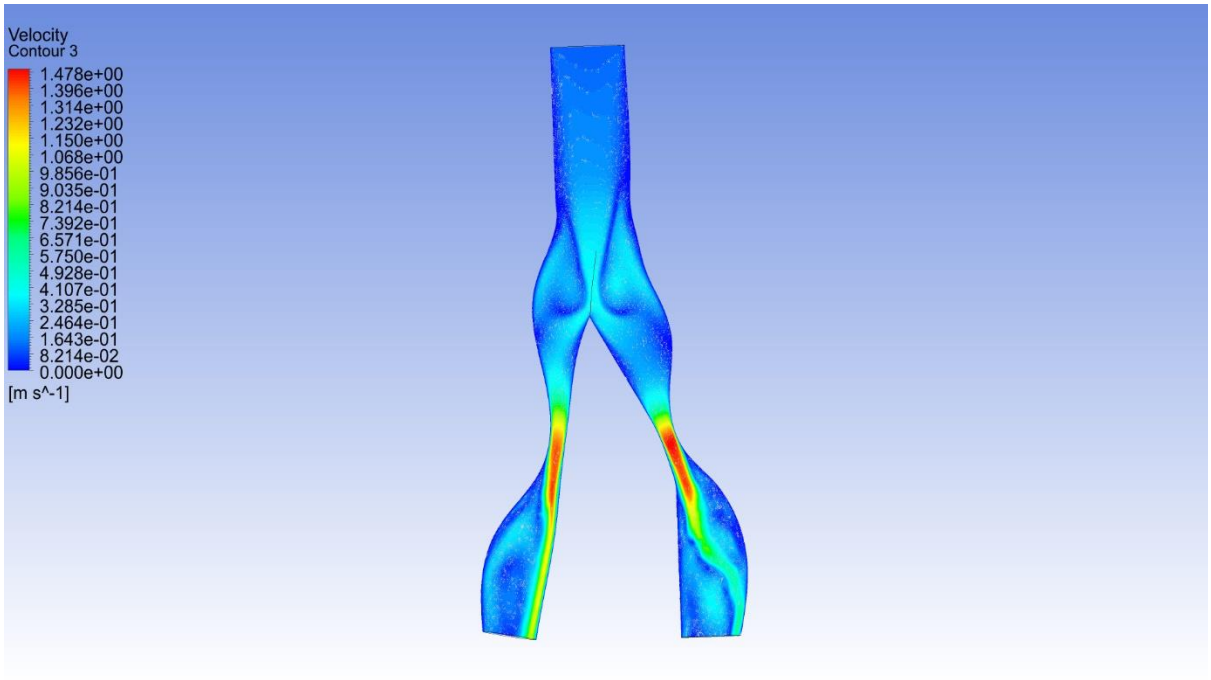
Fig. 4.5.3: Pressure vs Position

Figure 4.5.3 shows pressure vs position curve for both Newtonian and non-Newtonian model with plaque. As it can be seen pressure pattern is much differ than the without plaque and at some point there is no pressure. this is the outlet. Maximum pressure occurred at plaque zone.

4.6 Compression of velocity between two blood models with plaque



(a)



(b)

Fig. 4.6.1: Velocity Contour (a) non-Newtonian (b) Newtonian

Figure 4.6.1 and 4.6.2 shows that maximum velocity occurs at plaque area and the value is 1.4 m/s for both models. This in the plaque zone maximum velocity occurs and this could be one of vital issue of vascular disease and rupture of artery.

After this end of this chapter it can be said that because of plaque a lot of changes in the shear stress, pressure and velocity and this changes are not minor. Though, difference between Newtonian and non-Newtonian is not much. But, with plaque it is significant and wall shear stress, pressure and velocity is almost double or triple times than the normal geometry. This type of plaque is dangerous for diseases, heart blockage and atherosclerosis diseases. After all, it is seen the effect of velocity profile in bifurcation and plaque zone, effect of wall shear stress and effect of pressure similarly. So, objective of this study is fulfilled.

CHAPTER V
Conclusion

In this present study numerical analysis of pulsatile blood flow has been done and wall shear stress, pressure and velocity are compared between Newtonian and non-Newtonian blood models. This simulation is carried out in ANSYS 2020. Major finding of this research is given below:

- There could a lot of blood models but their difference is not much as it is seen in pressure and wall shear stress plot.
- It is seen that maximum pressure and wall shear stress is developed in the bifurcation zone.
- it can be said where is plaque there is more shear stress and pressure are likely to form. So, various vascular and atherosclerosis diseases is like to happen there.
- One of the most beneficial is that the results also may be of great importance for vessel surgery as grafts and artificial heart valves whose long-term performance still suffers from blood clot formation and thromboembolic events.
- From the comparison of model with plaque and not plaque it is seen that wall shear stress, pressure and velocity is more dangerous for artery has plaque in their artery.
- Clearly seen in the result and discussion chapter that every hemodynamics parameter is increased in the model with plaque than the model without plaque.
- hemodynamics parameter is likely to occur with maximum effect at the bifurcation zone and plaque area.
- Also, it is determined that all blood viscosity models are almost same with the value and they follow same trend of the graph.
- It can also be said that these hemodynamics parameters such as WSS, pressure and velocity are main reasons of vascular diseases.

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